

- 1) Mr. and Mrs. Jackson invested \$122,500 that they received from the proceeds for the sale of their house into a savings plan that pays 3.5% interest *compounded annually*. The Jacksons intend to leave all of the money in the savings plan for 6 years. How much will their investment be worth at the end of the 6-year period? Find the amount to the *nearest cent*.

- 2) Max will purchase a certificate of deposit (CD) for \$12,000. The CD pays 2.8% interest *compounded annually*. Jack has \$12,000 that he will invest in a plan that pays 3% *simple interest*. Which person's investment will be worth more money after 10 years if neither person withdraws any money from his investment?

- 3) Principal = \$25,750
Rate = 4 ½ % APR compounded annually
Time = 8 years
 - a. Find the value of the investment at the end of the investment period.
 - b. Determine the total amount of interest earned over the investment period.

- 4) Mr. Harris wrote the following formula for his algebra class to represent the value of a savings account over time (t) in years.

$$f(t) = (1800)(0.025)t + 1800$$

- a. What is the interest rate in Mr. Harris' formula? Give your answer as a percent.
- b. Is Mr. Harris' formula a simple interest formula or a compound interest formula?
- c. What does 1800 represent in Mr. Harris' formula?
- d. Use the formula to find $f(5)$.
- e. What does your answer to Part d above mean?
- f. Describe the graph of Mr. Harris' formula.

- 5) Ms. Sanchez wrote the following formula for her algebra class to represent the future value of an investment at any rate of interest (r).

$$FV = 14,500(1 + r)^{10}$$

- a. What does 14,500 represent in Ms. Sanchez's formula?
- b. What does 10 represent in Ms. Sanchez's formula?
- c. Is Ms. Sanchez's formula a simple interest formula or a compound interest formula?
- d. Use Ms. Sanchez's formula to determine the future value of the investment if interest rate is 8%. Give your answer to the *nearest cent*.